Skin Lesion Classification for Monkeypox Detection Using Handcrafted and Convolutional Neural Network Features

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Abstract—Monkeypox (MPX) is a zoonotic disease that presents with skin lesions visually similar to other conditions such as chickenpox and measles. Accurate automated classification of skin lesions is critical for timely diagnosis and containment. In this study, we propose a hybrid pattern recognition approach that combines handcrafted features and convolutional neural network (CNN) embeddings for binary classification of Monkeypox versus non-Monkeypox (ETC) skin lesions. Handcrafted features include color moments, Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP), Gabor filters, Hu moments, edge statistics, and Fast Fourier Transform (FFT) power spectrum features. CNN features are extracted from a pre-trained ResNet18 model. We evaluate four classifiers—Support Vector Machine (SVM), Random Forest, XGBoost, and Logistic Regression—on four feature representations: handcrafted only, CNN only, hybrid selected, and hybrid full. Experiments on a dataset of 228 skin lesion images (102 MPX and 126 ETC) demonstrate that the CNN-only feature representation with an SVM (RBF kernel) achieves the best performance, with a test accuracy of 85.5% and an AUC of 0.949. The hybrid approach also shows competitive results, with the full hybrid achieving 82.6% accuracy and 0.928 AUC. These findings suggest that transfer-learned CNN embeddings are highly effective for this classification task, while handcrafted features provide complementary information that can be leveraged when computational resources are limited.

Index Terms—Monkeypox classification, skin lesion, handcrafted features, GLCM, Local Binary Patterns, Gabor filters, ResNet18, transfer learning, pattern recognition

I. Introduction

Monkeypox (MPX) is a viral zoonotic disease caused by the Monkeypox virus, a member of the Orthopoxvirus genus. Although historically endemic to Central and West Africa, the 2022–2023 global outbreak highlighted the urgent need for rapid diagnostic tools. Clinical diagnosis of MPX relies heavily on the visual inspection of skin lesions, which can be easily confused with other vesicular or pustular diseases such as chickenpox, measles, and various skin rashes [1]. Automated image-based classification systems offer a promising avenue for assisting dermatologists and primary care physicians in resource-limited settings.

Pattern recognition and machine learning have been widely applied to medical image analysis, particularly for skin lesion classification [2]. Early approaches relied predominantly on handcrafted features—carefully designed descriptors that capture color, texture, and shape information from images. More recently, deep learning models, especially Convolutional Neural Networks (CNNs), have achieved state-of-the-art performance by learning hierarchical representations directly from raw pixels. However, CNNs typically require large datasets and significant computational resources. For smaller datasets, combining handcrafted features with CNN embeddings (hybrid approaches) has shown promise, leveraging the interpretability of handcrafted features alongside the discriminative power of deep representations [3].

In this work, we present a comprehensive pattern recognition pipeline for binary classification of Monkeypox versus non-Monkeypox skin lesions. Our contributions are threefold: (1) we implement a robust lesion segmentation pipeline based on the LAB color space and Otsu thresholding; (2) we extract a rich set of handcrafted features covering color, texture, shape, edge, and frequency domains; and (3) we systematically compare the performance of handcrafted features, CNN features from a pre-trained ResNet18, and their hybrid combinations across four different classifiers.

II. Related Work

A. Handcrafted Features for Skin Lesion Analysis

Handcrafted features have long been the backbone of medical image analysis. For skin lesion classification, color-based features such as RGB and HSV histograms and moments are commonly used because skin lesions often exhibit distinct color patterns [?]. Texture features are equally important, as the surface characteristics of lesions (e.g., smooth, rough, pustular) carry diagnostic information. The Gray-Level Co-occurrence Matrix (GLCM), introduced by Haralick et al. [4], is a classical method for quantifying texture through second-order statistical measures such as contrast, correlation, energy, and homogeneity. Local Binary Patterns (LBP), proposed by Ojala et al. [5], encode local texture patterns and have been successfully applied to skin lesion classification due to their invariance to monotonic illumination changes.

Gabor filters, which decompose images into components at multiple scales and orientations, are effective at capturing directional texture patterns in lesions [6].

Shape descriptors such as Hu moments provide rotation-, scale-, and translation-invariant characterizations of lesion morphology [7]. Combined with edge statistics (e.g., Canny edge density, gradient magnitude) and frequency-domain features from the Fast Fourier Transform (FFT), these handcrafted descriptors form a comprehensive representation of lesion appearance.

B. Deep Learning and Transfer Learning

Deep CNNs have revolutionized medical image classification. Models such as ResNet [8], VGGNet [9], and EfficientNet [10] learn hierarchical feature representations that are highly effective for visual recognition tasks. Transfer learning—fine-tuning or using as feature extractors pretrained models on large datasets such as ImageNet—has become a standard practice in medical imaging, where labeled data is often scarce [11]. For skin lesion classification, transfer-learned CNN features have consistently outperformed handcrafted feature-based methods on large benchmark datasets such as ISIC [2].

C. Hybrid Approaches

Several studies have explored hybrid feature representations that combine handcrafted and deep features. Ozkan and Dana [3] demonstrated that concatenating GLCM and LBP features with CNN embeddings improved classification accuracy over using either alone for melanoma detection. Similarly, Adegun and Viriri [12] showed that ensemble methods combining handcrafted features (color, texture, shape) with deep features achieved superior performance on dermoscopy images. These findings motivate our investigation into hybrid representations for Monkeypox classification.

III. Methodology

A. Dataset

The dataset used in this study comprises 228 skin lesion images, obtained from publicly available Monkeypox image collections. The dataset is divided into two classes: Monkeypox (MPX) with 102 images and Non-Monkeypox (ETC) with 126 images. All images are resized to 224×224 pixels with three color channels (RGB). The class distribution is slightly imbalanced ($\sim$ 45% MPX, 55% ETC), which is representative of real-world diagnostic scenarios. We use a stratified 70/30 train-test split, resulting in 160 training images and 68 test images, preserving the original class proportions.

B. Image Preprocessing and Lesion Segmentation

Accurate segmentation of the lesion from surrounding skin is critical for extracting meaningful features. We implement a segmentation pipeline based on the CIELAB color space, which separates luminosity (L^*) from color information (a^*, b^*). The a^* channel, which captures the red-green component, is particularly useful for skin lesions that often appear reddish or purplish relative to normal skin.

The segmentation process consists of the following steps:

1) Convert the input image from BGR to LAB color space.
2) Apply Contrast Limited Adaptive Histogram Equalization (CLAHE) to the L^* and a^* channels to enhance local contrast.
3) Combine the CLAHE-enhanced L^* and inverted a^* channels with weights 0.3 and 0.7, respectively, to produce a combined grayscale map.
4) Apply Otsu's thresholding to binarize the combined map.
5) Perform morphological closing (to fill holes) and opening (to remove noise) using an elliptical structuring element of size 5×5.
6) Retain only the largest connected component to isolate the primary lesion and eliminate spurious regions.

The resulting binary mask is applied to the original image to obtain the segmented lesion, which is then used for subsequent feature extraction.

C. Handcrafted Feature Extraction

We extract five categories of handcrafted features, yielding a total of approximately 200 raw features. After removing near-zero-variance features, 155 handcrafted features remain. Table I summarizes the feature categories.

TABLE I
Handcrafted Feature Categories

Category	Features	Count
Color	Moments (mean, std, CV) in BGR, HSV, LAB	27
	HSV histograms (16 bins $\times$ 3 channels)	48
	HSV percentiles (5^{th}, 50^{th}, 95^{th})	9
Texture (GLCM)	Contrast, dissimilarity, homogeneity, energy, correlation (mean + std over 4 angles, 2 distances)	10
Texture (LBP)	Uniform LBP histogram (radius=2, P=16)	18
Texture (Gabor)	Mean and std of Gabor magnitude at 3 orientations $\times$ 2 scales	12
Shape	Hu moments (7 invariants)	7
	Circularity, aspect ratio, extent, solidity, eccentricity, major/minor axis lengths	5
Edge	Canny edge density	1
	Gradient mean and std (Sobel)	2
Frequency	FFT radial power spectrum mean and std (8 ring bands)	16

Color Features. For each color space (BGR, HSV, LAB), we compute the mean, standard deviation, and coefficient of variation (CV) for each channel within the masked lesion region. Additionally, we compute 16-bin normalized histograms for each HSV channel and extract the 5^{th},

50^{th}, and 95^{th} percentiles to capture color distribution characteristics.

GLCM Features. The Gray-Level Co-occurrence Matrix captures spatial relationships between pixel intensities. We quantize grayscale images to 64 levels and compute GLCMs at distances $d \in \{1, 2\}$ and angles $\theta \in \{0°, 45°, 90°, 135°\}$. From each GLCM, we extract five Haralick properties: contrast, dissimilarity, homogeneity, energy, and correlation. The mean and standard deviation across all orientations and distances are computed to produce rotation-invariant descriptors.

LBP Features. Local Binary Patterns encode local texture by comparing each pixel with its circularly symmetric neighbors. We use uniform LBP with radius $r = 2$ and $P = 16$ sampling points, resulting in 18 histogram bins. The histogram is normalized to serve as a texture descriptor invariant to monotonic illumination changes.

Gabor Features. Gabor filters decompose images into components at different spatial frequencies and orientations. We apply a bank of filters at 3 orientations $(0°, 60°, 120°)$ and 2 scales $(\sigma \in \{1.5, 3.0\})$, and compute the mean and standard deviation of the response magnitude within the lesion mask for each filter.

Shape Features. From the binary segmentation mask, we compute Hu moments, which are invariant to translation, rotation, and scaling. We also derive geometric properties from the largest contour: circularity, aspect ratio, extent, solidity, eccentricity, and major/minor axis lengths.

Edge Features. Edge density is computed as the proportion of Canny-detected edge pixels within the image. Gradient statistics (mean and standard deviation of Sobel gradient magnitude) within the lesion region capture border sharpness and texture roughness.

FFT Features. The 2D Fast Fourier Transform decomposes the image into frequency components. We compute the mean and standard deviation of the magnitude spectrum within 8 concentric ring bands centered at the origin, capturing frequency-domain energy distribution.

D. CNN Feature Extraction

To capture high-level semantic representations, we employ a pre-trained ResNet18 model [8] as a fixed feature extractor. The model was pre-trained on the ImageNet dataset (1.2 million images, 1,000 classes). We remove the final fully connected (classification) layer and use the output of the global average pooling layer, which produces a 512-dimensional feature vector for each input image.

Input images are preprocessed by resizing to 224×224 pixels, converting to PyTorch tensors, and normalizing using the ImageNet mean $([0.485, 0.456, 0.406])$ and standard deviation $([0.229, 0.224, 0.225])$. Feature extraction is performed in inference mode (no gradient computation) on the available hardware (CPU or CUDA GPU).

E. Feature Fusion and Selection

We investigate four feature representations:

1) Handcrafted only (155 features after variance filtering).
2) CNN only (512-dim ResNet18 embeddings).
3) Hybrid (full): concatenation of handcrafted and CNN features (667 dimensions).
4) Hybrid (selected): the top 200 features from the full hybrid set selected by mutual information with the class labels.

All features are standardized using Z-score normalization ($\mu = 0$, $\sigma = 1$) prior to classification.

F. Classification Algorithms

We evaluate four widely used classifiers:

- Support Vector Machine (SVM) with RBF kernel ($C = 10$, $\gamma = \text{scale}$).
- Random Forest (200 estimators, max depth 10).
- XGBoost (200 estimators, max depth 6, learning rate 0.1).
- Logistic Regression ($C = 1.0$, max iterations 5000).

All hyperparameters are fixed across feature sets to ensure fair comparison. We report 5-fold stratified cross-validation metrics on the training set and independent test set performance.

IV. Experimental Setup and Results

A. Implementation Details

The pipeline is implemented in Python 3.12 using OpenCV for image processing, scikit-image for texture analysis, PyTorch for CNN feature extraction, and scikit-learn for classification. The experiments are conducted on a machine with an NVIDIA GPU (CUDA 12.1). Feature extraction is cached to disk to enable rapid experimentation. The random seed is fixed at 42 for reproducibility.

B. Segmentation Qualitative Results

Figure 1 shows examples of the segmentation pipeline applied to sample images from both classes. The LAB-based Otsu thresholding effectively isolates the primary lesion region while suppressing background skin and hair artifacts.

C. Classification Results

Table II presents the test-set performance of all classifier–feature-set combinations, sorted by test AUC. The best overall performance is achieved by the SVM with RBF kernel using CNN-only features, attaining a test accuracy of 85.5% and an AUC of 0.949.

The CNN-only representation consistently outperforms the handcrafted-only baseline across all classifiers. Notably, the hybrid selected representation (200 features) achieves strong performance while being more compact than the full hybrid, reducing the risk of overfitting on this small dataset.

The classification report for the best-performing model (SVM + CNN) is shown in Table III.

TABLE II
Test Set Performance Across Feature Sets and Classifiers (sorted by Test AUC, descending)

Feature Set	Classifier	Test Acc	Test F1 (macro)	Test AUC	CV AUC
CNN (ResNet18)	SVM (RBF)	0.855	0.853	0.949	0.891
Hybrid (full)	XGBoost	0.841	0.838	0.890	0.798
Hybrid (selected)	XGBoost	0.855	0.848	0.907	0.833
Hybrid (selected)	SVM (RBF)	0.841	0.840	0.923	0.916
CNN (ResNet18)	Random Forest	0.812	0.806	0.911	0.821
CNN (ResNet18)	Logistic Regression	0.826	0.823	0.887	0.847
Hybrid (full)	SVM (RBF)	0.826	0.825	0.928	0.915
Hybrid (full)	Logistic Regression	0.812	0.808	0.885	0.896
Handcrafted	XGBoost	0.754	0.752	0.840	0.768
CNN (ResNet18)	XGBoost	0.797	0.794	0.869	0.805
Handcrafted	Random Forest	0.739	0.735	0.784	0.794
Handcrafted	Logistic Regression	0.667	0.666	0.778	0.763
Handcrafted	SVM (RBF)	0.594	0.592	0.711	0.776

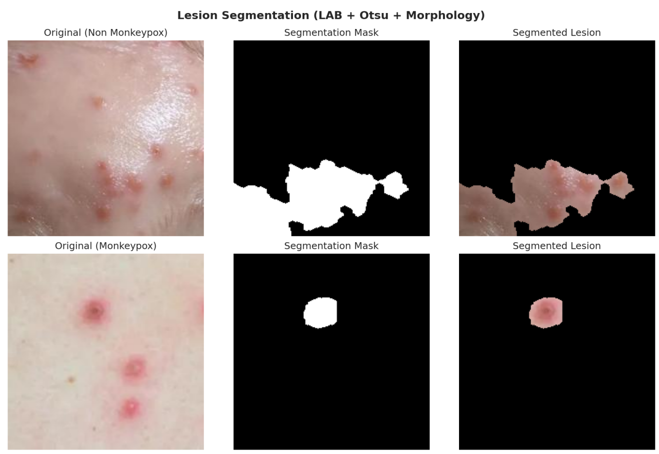

Fig. 1. Examples of lesion segmentation. For each row: (left) original image, (middle) binary mask, (right) segmented lesion overlay.

TABLE III
Classification Report for Best Model (SVM + CNN-only)

Class	Precision	Recall	F1-Score
Monkeypox	0.86	0.81	0.83
Non-Monkeypox	0.85	0.89	0.87
Macro Avg	0.86	0.85	0.85
Weighted Avg	0.86	0.86	0.85

D. Visualization and Analysis

ROC Curves. Figure 2 compares the ROC curves for the SVM classifier across the four feature representations. The CNN-only curve dominates the others, reflecting its superior discriminative power.

Confusion Matrix. Figure 3 shows the confusion matrix for the best model. The model correctly identifies 81% of Monkeypox cases and 89% of non-Monkeypox cases. Misclassifications are roughly balanced between false positives and false negatives.

Feature Space Visualization. Figure 4 shows a t-SNE projection of the hybrid (selected) feature space colored

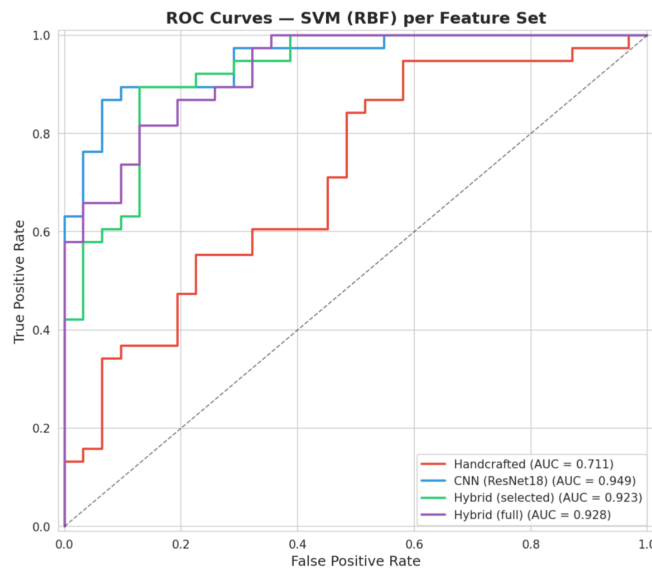

Fig. 2. ROC curves for SVM (RBF) across feature sets. The CNN-only representation achieves the highest AUC (0.949).

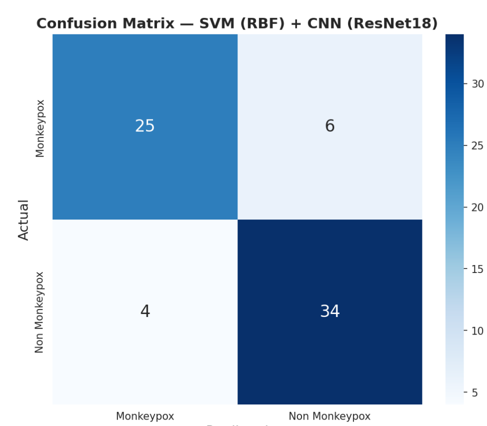

Fig. 3. Confusion matrix for the best-performing model (SVM + CNN-only features).

by class label. A moderate degree of class separation is visible, although some overlap remains, consistent with the challenging nature of visual MPX diagnosis.

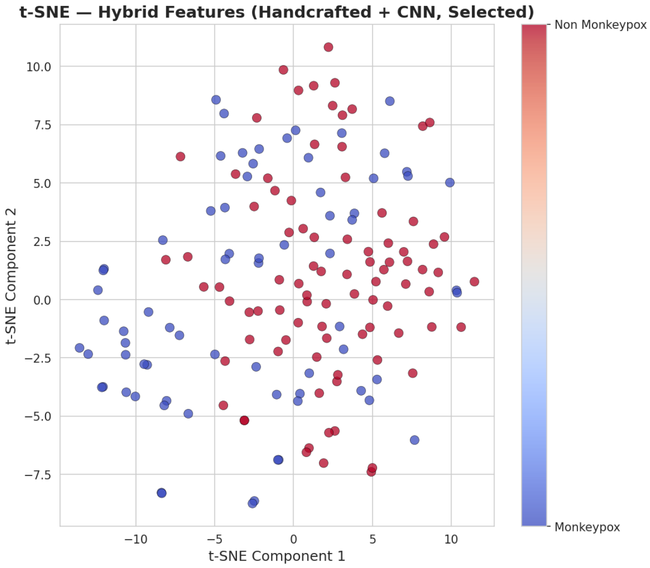

Fig. 4. t-SNE visualization of the hybrid (selected) feature space on the training set. Classes show moderate separability.

Feature Importance. Figure 5 shows the cumulative importance of handcrafted feature categories according to a Random Forest classifier trained on handcrafted features alone. Color and texture (GLCM) features contribute most to the classification decision, followed by Gabor and edge features.

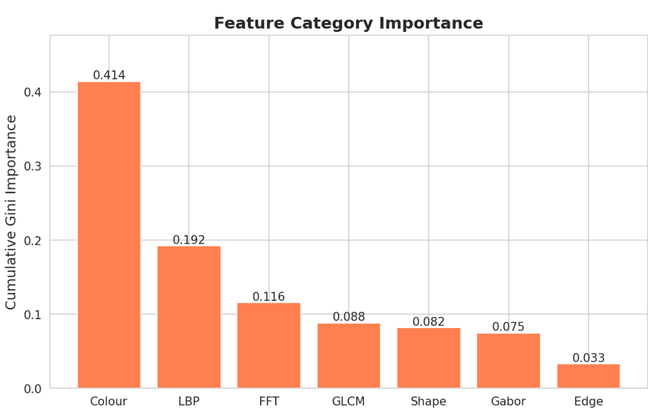

Fig. 5. Cumulative Gini importance of handcrafted feature categories (Random Forest on handcrafted-only features).

V. Discussion

Our experiments reveal several important insights. First, CNN features extracted from a pre-trained ResNet18 are remarkably effective for Monkeypox classification, even without fine-tuning on the target dataset. This confirms the power of transfer learning for medical image analysis with limited data. The 512-dimensional embeddings capture high-level visual patterns—such as lesion shape, color distribution, and surface texture—that are informative for distinguishing MPX from other skin conditions.

Second, handcrafted features alone achieve moderate performance (best AUC 0.840 with XGBoost), suggesting that the designed descriptors capture clinically relevant information. The most important handcrafted categories are color and GLCM texture, which aligns with dermatological knowledge that MPX lesions have characteristic red-purple hues and specific surface textures (e.g., umbilicated vesicles).

Third, the hybrid approach does not surpass the CNN-only baseline on this dataset. We attribute this to the small sample size (228 images): the 667-dimensional full hybrid representation risks overfitting, despite feature selection reducing it to 200 dimensions. With a larger dataset, the hybrid approach might outperform either individual representation, as the handcrafted features could provide complementary information not captured by the generic ImageNet-trained CNN.

Finally, the choice of classifier matters. SVM with RBF kernel consistently performs best across all feature sets, likely due to its ability to model non-linear decision boundaries in high-dimensional spaces. Random Forest and XGBoost are also competitive, while Logistic Regression—a linear model—lags slightly, suggesting that the optimal decision boundary is non-linear.

VI. Conclusion

In this study, we presented a comprehensive pattern recognition pipeline for automated classification of Monkeypox versus non-Monkeypox skin lesions. We implemented robust lesion segmentation, extracted rich handcrafted features spanning color, texture, shape, edge, and frequency domains, and combined them with transfer-learned CNN embeddings from ResNet18. Our experiments on a dataset of 228 images demonstrate that CNN-only features with an SVM classifier achieve the best performance (85.5% accuracy, 0.949 AUC), outperforming both handcrafted-only and hybrid representations.

Future work will focus on: (1) collecting a larger, multi-center dataset to enable fine-tuning of deep models and more robust evaluation of hybrid approaches; (2) integrating attention mechanisms to highlight discriminative lesion regions; and (3) deploying the model as a lightweight mobile application for point-of-care diagnostic assistance in endemic regions.

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